



SMART FOOD DONATION PLATFORM

TEAM NAME:

Vision Vault

MENTOR DETAILS:

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PROBLEM VALIDATION & SDG VALIDATION

Problem Statement:

Large quantities of edible food are waste d daily while many people suffer from hunger due to lack of coordination between food donors and organizations.

Why This Problem Matters:

- Food waste causes environmental damage
- Millions face food insecurity
- No etficient redistribution system

Target Users:

- Restaurants, bakeries, superm arkets
- NGOs, orphanages, old age homes
- Volunteers

Gap in Existing Systems:

- No centralized platform
- No real-time coordination
- No tracking or transparency

SDG Alignment:

Supports SDG 2 - Zero Hunger





PROPOSED SOLUTION OVERVIEW

Proposed Solution:

We propose Smart Food Donation Platform a web-based platform that connects surplus food providers with volunteers and NGOs to ensure food reaches people in need before it expires

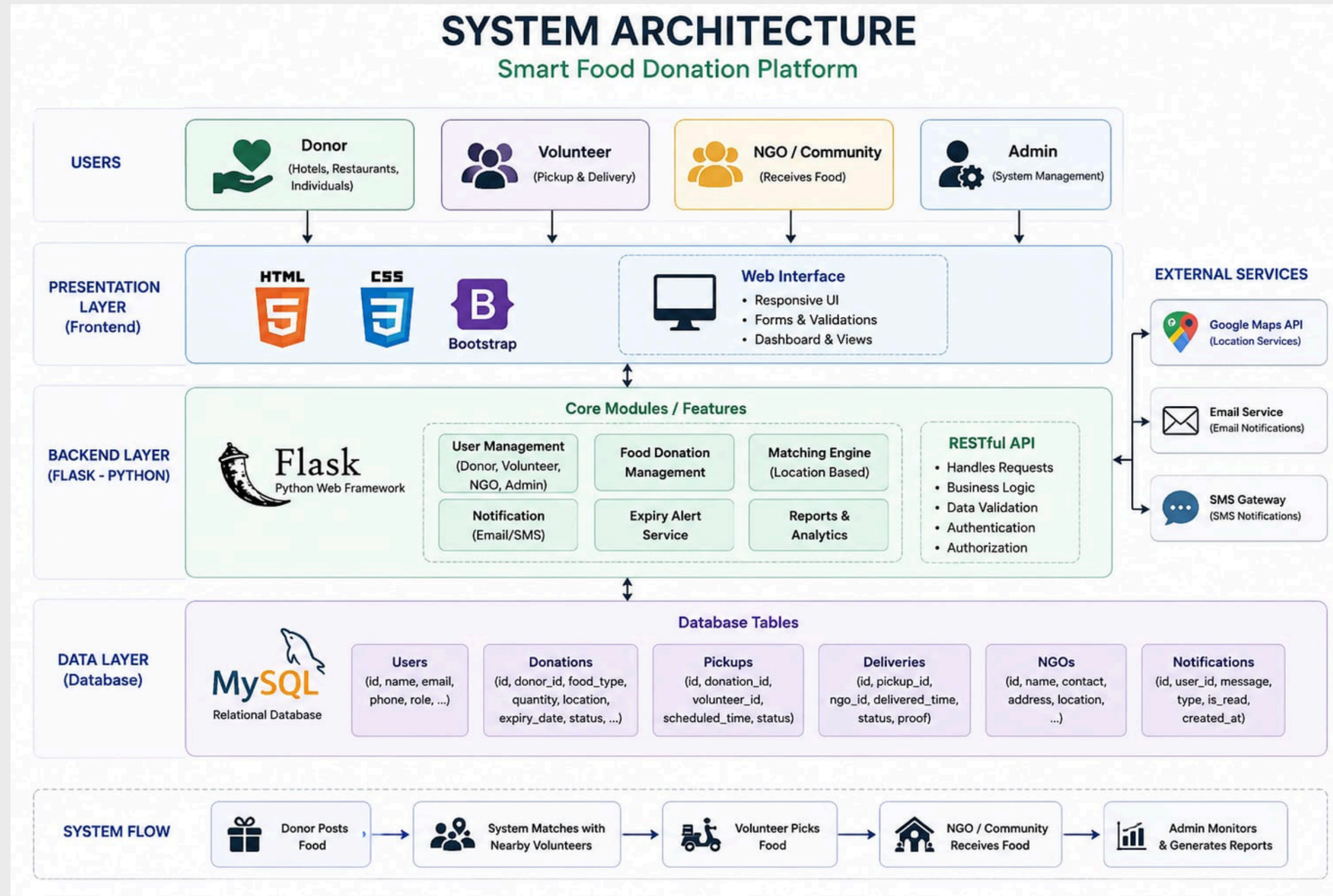
Key Features

- Donor dashboard to post food donations
- Volunteer dashboard to accept pickup requests
- NGO dashboard to receive and track food distribution
- Admin dashboard to monitor the system

Smart Features:

- Location-based matching of donors and volunteers
 - Food expiry priority alerts
 - Hunger heatmap to identify high-demand area
 - Secure verification for donors and NGOs
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SYSTEM ARCHITECTURE





METHODOLOGY & ALGORITHM

Workflow:

- Donor registers and posts food
- System identifies nearby volunteers
- Volunteer accepts request
- Volunteer collects and delivers food
- NGO confirms delivery
- Admin monitors activity

Algorithms /Logic:

- Location-based matching
- Expiry-based priority
- Role -based access control

Modules Developed:

- Donor dashboard
 - Volunteer dashboard
 - NGO dashboard
 - Admin dashboard
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TOOLS & TECHNOLOGIES USED

Frontend (User Interface)

Technology:

HTML, CSS, Bootstrap

Backend (Server) Technology:

Flask (Python)

Database Technology:

MySQL

APIs and Integrations:

GoogleMaps API

- show donor location
- navigation for volunteers
- map-based food pickup

Development Tools:

- Tools used to build the project.
 - Visual Studio Code
 - GitHub (version control)
 - Thunder Client (API testing)
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RESULT & PERFORMANCE ANALYSIS

Outputs:

- Functional dashboards
- Food donation workflow

Performance:

- Faster food distribution
- Reduced manual effort
- Real-time tracking

Innovation:

- Smart food redistribution
- Role-based system
- Real-time tracking
- Heatmap analysis

Impact:

- Reduces food waste
 - Helps needy people
 - Supports NGOs
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FUTURE WORK

1.AI-Based Smart Prediction System:

- Use Machine Learning to predict food dem and in different areas
- Suggest donors where food is most needed
- Reduce wastage even before it happens

2.Real-Time GPS Tracking System:

- Track volunteers during food delivery
- Provide live location updates to donors and NGOs
- Improve delivery transparency and efficiency

3.Mobile Application Development:

- Build Android & iOS apps for easy access
 - Push notifications for instant updates
 - Better user engagement
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REFERENCES

- United Nations - SDG 2: Zero Hunger
 - Food and Agriculture Organization (FAO) - Food Waste Statistics
 - HTML, CSS, Bootstrap Documentation
 - MY SQL Documentation
 - Google Maps API Documentation
 - Research papers on food waste management (IEEE I Springer)
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